Cryptography

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1 Introduction

Cryptography has always been about **message confidentiality** — making sure that a message can only be understood by its sender and intended recipient. It typically involves converting a message into an incomprehensible format, and then that process being reversed by the recipient. A good cryptography system should leave a potential eavesdropper feeling helpless

— they should feel like they don't have the time or energy to figure out what the message says.

Of course, the standard for "time or energy" has changed drastically with computers. What seemed like an unbreakable cipher to someone working by hand in the 1800s can now be cracked in milliseconds with a short script. But computers cut both ways — they also allow us to *create* far more complex encryption systems than any human could manage by hand.

Even with computers, some categories of problems remain genuinely hard. Recall how the Towers of Hanoi (from CSC 220 and CSC/CYEN 131) becomes intractable very quickly, even though the algorithm itself is simple to write. A good cryptography system might be easy to *understand* — you can explain exactly how it works — but should still leave even a powerful computer without enough time or memory to recover the original message.

Key Definitions

Term	Definition
Cryptology	The overall field: enciphering and deciphering
Cryptography	The art of *making* a cipher system
Cryptanalysis	The art of *breaking* a cipher system
Encryption	Scrambling a message to hide its meaning
Decryption	Unscrambling an encrypted message
Plain text	The original, human-readable message
Cipher text	The encrypted, unreadable message
Cipher	The algorithm used to perform encryption or decryption
Key	The secret value that controls how a cipher encrypts or decrypts

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2 A History of Cryptography

Classical cryptography was done with pen, paper, and occasionally simple mechanical tools. As you read through each cipher below, think about two questions:

1. **How would you break it?** What is the weakness an attacker could exploit?
2. **How hard would it be with modern technology?** Could a script crack it in under a second, or would it take longer?

The highlighted yellow passages are cipher text challenges for you to try and decrypt.

2.1 Hieroglyphs *(~3200 BC – 400 AD)*

Ancient Egyptians used images and symbols to represent messages — but only those initiated into the meaning of the symbols could interpret them. In a sense this is no different from writing in any natural language: the "secrecy" comes entirely from obscurity. The more people who know the language, the less effective it is at keeping messages secret.

> How Would You Break It?
>
> Hieroglyphs aren't really a cipher — they're a writing system. Breaking it just means learning the language, or finding a bilingual key (like the Rosetta Stone, which let historians decode Egyptian hieroglyphs by comparing the same text written in Greek). Modern computers could be trained to recognise hieroglyph patterns statistically, but the "key" here is really just knowledge of the language itself.

2.2 Atbash Cipher *(~500 BC – 1300 AD)*

Atbash is a **substitution cipher** where the alphabet is simply reversed: A becomes Z, B becomes Y, C becomes X, and so on. It is symmetric — applying it twice returns the original message. Examples of Atbash can actually be found in the Bible [1].

rhm'g xibkgltizksb ufm?

You can apply and reverse Atbash on the command line:

```
# Create the reversed alphabet z through a
echo {z..a} | tr -d ' '

# Encode or decode a string using Atbash (it's its own inverse)
echo "abcde" | tr a-z "zyxwvutsrqponmlkjihgfedcba"
```

[1] See: https://www.theology.ox.ac.uk/article/crack-the-code

How Would You Break It?

There is only one possible Atbash mapping — there is no key. If you know or suspect the cipher is Atbash, decryption is instant. Even if you don't know, the small number of substitution options makes it trivially brute-forceable. A modern script breaks it in microseconds.

2.3 Scytale Cipher *(~7th Century BC)*

The Scytale was a physical encryption device used by the ancient Spartans. A strip of parchment was wound around a cylindrical rod, and the message was written along the length of the rod. When the strip was unwound, the letters appeared in a scrambled order — a **transposition cipher**, meaning the letters are rearranged rather than substituted. Only a rod of exactly the same diameter could reconstruct the message.

Sdeeeiootyrdnnmhdbwgzeareatostaithoattncimihhbhs

How Would You Break It?

The "key" here is the diameter of the rod — which controls how many columns the text is split into. A modern attacker simply tries every possible column width (there are only as many options as the message is long), reads across each, and checks if the result is legible English. A script can brute-force every possibility in milliseconds.

2.4 Caesar Cipher *(~60 BC)*

The Caesar cipher is another substitution cipher, named after Julius Caesar who reportedly used it in his personal correspondence. Every letter in the message is replaced by the letter a fixed number of positions further along in the alphabet, wrapping around at the end.

Caesar himself shifted by 3 positions: A→D, B→E, C→F, ..., X→A, Y→B, Z→C. This specific version is called **ROT-3**. The most well-known variant today is **ROT-13**, which shifts by 13 — and because 13 is exactly half of 26, applying it twice returns the original message (it is its own inverse).

diwtgh hpxs: durdjght xi xh

```
echo "abcde" | tr a-z d-za-c   # ROT-3 (Caesar's cipher)
echo "abcde" | tr a-z n-za-m   # ROT-13
```

How Would You Break It?

There are only 25 possible shifts (ROT-1 through ROT-25). A human can check all of them by eye in a few minutes. A script can check all 25 in microseconds and automatically flag whichever shift produces readable English using a dictionary check. The Caesar cipher offers essentially zero security against modern computing.

2.5 Sliding Shift Cipher

The Caesar cipher's weakness is that every letter uses the *same* shift, which makes it easy to check all possibilities. The sliding shift cipher tries to fix this by using a *different* shift for each position in the message: the first letter is shifted by 1 (ROT-1), the second by 2 (ROT-2), the third by 3 (ROT-3), and so on.

This increases the number of possible combinations dramatically compared to Caesar — but not as much as it might seem.

How Would You Break It?

If an attacker knows the cipher is a sliding shift (or suspects it), the starting offset is the only unknown. There are only 25 possible starting offsets, making it barely harder to brute-force than a plain Caesar cipher. Even without knowing the cipher type, statistical analysis of letter frequencies across positions would reveal the shifting pattern. This cipher is a stepping stone to something more powerful — the Vigenère cipher.

2.6 Vigenère Cipher *(~1553 – 1863)*

The Vigenère cipher generalises the sliding shift idea by using a **repeating keyword** as the key, rather than a simple incrementing counter. Each letter of the plain text is shifted by an amount determined by the corresponding letter of the key. When the key runs out, it repeats from the beginning.

For example, with the key `"vigenere"` and the plain text `"how does this work"`:

Table 2: Vigenère encryption example

Plain text	H	O	W	D	O	E	S	T	H	I	S	W	O	R	K
Key	V	I	G	E	N	E	R	E	V	I	G	E	N	E	R
Cipher text	C	W	C	H	B	I	J	X	C	Q	Y	A	B	V	B

The key letter determines the row in the Vigenère table, and the plain text letter determines the column. Where they intersect is the cipher text character.

	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
a:	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
b:	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a
c:	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b
d:	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c
e:	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d
f:	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e
g:	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f
h:	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g
i:	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h
j:	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i
k:	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j
l:	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k
m:	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l
n:	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m
o:	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n
p:	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
q:	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p
r:	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q
s:	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r
t:	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s
u:	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t
v:	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u
w:	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v
x:	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w
y:	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x
z:	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y

Figure 1: Vigenère Chart

To **encrypt**: find the plain text letter's column and the key letter's row — the intersection is your cipher text. For example, plain text **l** with key **n** gives cipher text **y**.

To **decrypt**: find the key letter's row, then find the cipher text character in that row. The column header it falls under is the plain text character. For example, cipher text **p** with key character **x** gives plain text **s**.

2.6.1 The Maths Behind It

Performing lookups in a table by hand is slow. Computers make this far faster using arithmetic. If we assign each letter a number (a=0, b=1, ..., z=25), the Vigenère cipher becomes simple modular addition.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

For example: plain text **l** has value 11, key character **n** has value 13. $(11 + 13)\%26 = 24$, which maps to **y** — the same result as the table lookup.

More formally, if P is the plain text, K the key, C the cipher text, and the subscript i the character at position i:

$$C_i = (P_i + K_i)\%26$$

To decrypt:

$$P_i = (26 + C_i - K_i)\%26$$

The 26 is added before subtracting to avoid negative numbers, since different programming languages handle the modulo of negative numbers inconsistently.

Worked Example

Plain text: **"cyberstorm is going to be bussin"** — Key: **"cryptids"**

Position	Plain	Key	Calculation	Cipher
0	c (2)	c (2)	(2+2)%26 = 4	**e**
1	y (24)	r (17)	(24+17)%26 = 15	**p**
2	b (1)	y (24)	(1+24)%26 = 25	**z**
3	e (4)	p (15)	(4+15)%26 = 19	**t**
4	r (17)	t (19)	(17+19)%26 = 10	**k**

5	s (18)	i (8)	(18+8)%26 = 0	**a**
6	t (19)	d (3)	(19+3)%26 = 22	**w**
…	…	…	…	…

Cipher text: **epztkawgtd gh zwlfi km qx jxkuzl**
To decrypt **"avq nhcu htfdtlarj wjcs ucvkke. Ft'l maltr. Ld vis."** with the same key:

Position	Cipher	Key	Calculation	Plain
0	a (0)	c (2)	(26+0−2)%26 = 24	**y**
1	v (21)	r (17)	(26+21−17)%26 = 4	**e**
2	q (16)	y (24)	(26+16−24)%26 = 18	**s**
3	n (13)	p (15)	(26+13−15)%26 = 24	**y**
4	h (7)	t (19)	(26+7−19)%26 = 14	**o**
5	c (2)	i (8)	(26+2−8)%26 = 20	**u**
6	u (20)	d (3)	(26+20−3)%26 = 17	**r**
…	…	…	…	…

Plain text: **yes your ...**

Characters as Numbers in Code

Since many programming languages store characters as integers internally, converting a character to its numerical value (0–25) is as simple as subtracting the ASCII value of `'a'`:

```python
# Python
ord('l') - ord('a')   # → 11
ord('n') - ord('a')   # → 13
```

```cpp
// C++
'P' - 'A'   // → 15  (80 - 65)
'p' - 'a'   // → 15  (112 - 97)
```

```go
// Go
// Characters (runes) are just integers, so arithmetic works directly.
// Single-quoted literals have type rune (alias for int32).
'l' - 'a'   // → 11  (108 - 97)
'n' - 'a'   // → 13  (110 - 97)
'P' - 'A'   // → 15  (80 - 65)

// In practice, when iterating over a string you cast to int explicitly:
ch := 'l'
val := int(ch) - int('a')   // → 11
```

Go's character type is called a **rune** (an alias for `int32`), which can represent any Unicode code point — not just ASCII. For the purposes of Vigenère encryption over the English alphabet, subtracting `'a'` works identically to Python and C++. The cast to `int` is needed when you want to do arithmetic and store the result as a plain integer rather than a rune.

How Would You Break It?

The Vigenère cipher was considered unbreakable for nearly 300 years — until Charles Babbage and Friedrich Kasiski independently cracked it in the 1800s using **frequency analysis** combined with **key length detection**.
The key insight is that if you can figure out the key length n, you can split the cipher text into n groups (all characters encrypted with the same key letter) and treat each group as a simple Caesar cipher — which is trivially crackable using English letter frequency statistics (e in English is the most common letter, followed by t, a, o, i, n…).
Key length detection: look for repeated sequences of characters in the cipher text. The distances between repetitions tend to be multiples of the key length. A modern script can determine key length and break the cipher in well under a second.

How Computers Changed Cryptography

Historically, cryptography was almost always about hiding written text. Computers changed the game in three fundamental ways:

1. **Speed** — Computers can attempt millions of decryptions per second, making short keys or small key spaces completely insecure.
2. **Complexity** — Computers can execute ciphers that would be impossibly slow to carry out by hand, enabling far more sophisticated algorithms.

3. **Scope** — Computers can encrypt *any* digital data — images, executables, video, databases — not just text. Anything representable as 1s and 0s can be encrypted. This is arguably the most transformative change.

---

3 Encoding

Cryptography is not only about *hiding* information. Sometimes it is simply about **representing** information in a format that is convenient to work with or transmit. These representations are called **encodings** — and unlike encryption, they typically involve no secret key and no intent to hide meaning.

> ⚠ Encoding Is Not Encryption
>
> This is a common source of confusion. Encoding (e.g. Base64) is a reversible transformation with a publicly known algorithm and no key. Anyone who knows the encoding scheme can reverse it instantly. **Encoded data is not secret.** Do not use Base64 as a substitute for encryption.

3.1 ASCII

ASCII (American Standard Code for Information Interchange) is the foundational encoding scheme for representing text in computers. It was originally 7-bit, allowing 128 possible characters — enough for uppercase and lowercase English letters, digits, punctuation, and control characters (like newline and tab). As 8-bit storage became standard, ASCII was extended to 256 characters.

Common values to know:

Character	Decimal	Binary
`A`	65	01000001
`B`	66	01000010
`a`	97	01100001
`b`	98	01100010
`0`	48	00110000
Space	32	00100000

Notice that lowercase letters have values exactly 32 higher than their uppercase counterparts — which is why flipping bit 5 (value 32) toggles between upper and lowercase in ASCII.

> **Why Does This Matter for Cryptography?**
>
> When we perform Vigenère encryption by mapping `'a'` → 0, `'b'` → 1, …, we are essentially working with ASCII values and subtracting the base offset. Understanding that characters *are* numbers at the hardware level is fundamental to implementing any cipher in code.

3.2 Base64

Base64 was created to solve a practical problem: how do you send arbitrary binary data (like an image or a file attachment) over a channel that was designed to carry only printable text? Early email systems, for example, could not reliably handle raw binary bytes — so Base64 was developed as a way to represent any binary stream using only 64 safe printable characters.

You can identify Base64-encoded data by its character set: only uppercase letters, lowercase letters, digits, `+`, and `/`, often with one or two `=` signs at the end.

3.2.1 How Base64 Works

The conversion process follows the same principle as converting binary to hexadecimal: group the bits, then look up each group in the encoding table. For Base64, the group size is **6 bits** (since $2^6 = 64$).

Table 6: Standard Base64 encoding table

Value	Char	Value	Char	Value	Char	Value	Char
0	A	16	Q	32	g	48	w
1	B	17	R	33	h	49	x
2	C	18	S	34	i	50	y
3	D	19	T	35	j	51	z
4	E	20	U	36	k	52	0
5	F	21	V	37	l	53	1
6	G	22	W	38	m	54	2
7	H	23	X	39	n	55	3
8	I	24	Y	40	o	56	4
9	J	25	Z	41	p	57	5
10	K	26	a	42	q	58	6
11	L	27	b	43	r	59	7

Value	Char	Value	Char	Value	Char	Value	Char
12	M	28	c	44	s	60	8
13	N	29	d	45	t	61	9
14	O	30	e	46	u	62	+
15	P	31	f	47	v	63	/

3.2.2 Worked Example: `"abc"` → Base64

Input	a								b								c							
ASCII	97								98								99							
Binary	0	1	1	0	0	0	0	1	0	1	1	0	0	0	1	0	0	1	1	0	0	0	1	1
Index	24								22				9				35							
Base-64	Y								W				J				j							

Take the ASCII bytes for `a`, `b`, `c` (01100001, 01100010, 01100011), concatenate the 24 bits, split into groups of 6, and look each up in the table. The result is **YWJj** — a perfect conversion since 3 bytes × 8 bits = 24 bits, which divides evenly by 6.

```
echo -n "abc" | base64   # -n suppresses the trailing newline
# Output: YWJj
```

3.2.3 Padding with =

When the number of input bytes is not a multiple of 3, the bit stream does not divide evenly into groups of 6. Padding bits are added to fill the last group, and `=` characters are appended to the output to signal that padding occurred.

Example: `"x"` → Base64

Input	x																							
ASCII	120								0								0							
Binary	0	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Index	30								0				0				0							
Base-64	e								A				=				=							

`x` is a single byte (8 bits). We need at least 12 bits to fill two 6-bit groups, so 4 padding bits are added. The second group now contains 2 real bits and 4 padding bits. The remaining

12-bit group is pure padding and cannot be meaningfully encoded, so it becomes `==` in the output. The result is **eA==**.

Example: `"CS"` → Base64

Input	C								S																			
ASCII	67								83								0							
Binary	0	1	0	0	0	0	1	1	0	1	0	1	0	0	1	1	0	0	0	0	0	0	0	0
Index	16				53				12				0											
Base-64	Q				1				M				=											

Two bytes = 16 bits. We need 18 bits for three 6-bit groups, so 2 padding bits are added. Only one `=` is needed. The result is **Q1M=**.

```
echo -n "CS" | base64
# Output: Q1M=
```

3.2.4 Decoding Base64

Example: `"RU4="` → ASCII

Base-64	R						U						4						=					
Index	17						20						56						0					
Binary	0	1	0	0	0	1	0	1	0	1	0	0	1	1	1	0	0	0	0	0	0	0	0	0
ASCII	69								78								0							
Output	E								N															

Strip the `=`, convert each Base64 character to its 6-bit value, concatenate the bits, remove the padding, and group into 8-bit ASCII bytes. The result is **EN**.

```
echo -n "RU4=" | base64 -d   # -d decodes Base64
# Output: EN
```

---

4 Cryptography Today

The internet created a need for cryptography far beyond simple message confidentiality. We now communicate over a network that no single entity controls, where bad actors can intercept traffic at many points. Modern cryptography must solve several distinct problems simultaneously:

- **Confidentiality** — only the intended recipient can read the message
- **Integrity** — the message has not been altered in transit
- **Authenticity** — the sender is who they claim to be
- **Non-repudiation** — the sender cannot later deny having sent the message

The three major tools used to address these problems are symmetric encryption, asymmetric encryption, and hashing.

4.1 Symmetric-Key Cryptography

Symmetric-key cryptography works like the classical ciphers above: both the sender and receiver share the same secret key, which is used for both encryption and decryption. Depending on the algorithm, data can be encrypted as a continuous stream (**stream ciphers**, one byte at a time) or in fixed-size chunks (**block ciphers**, where each block is encrypted as a unit).

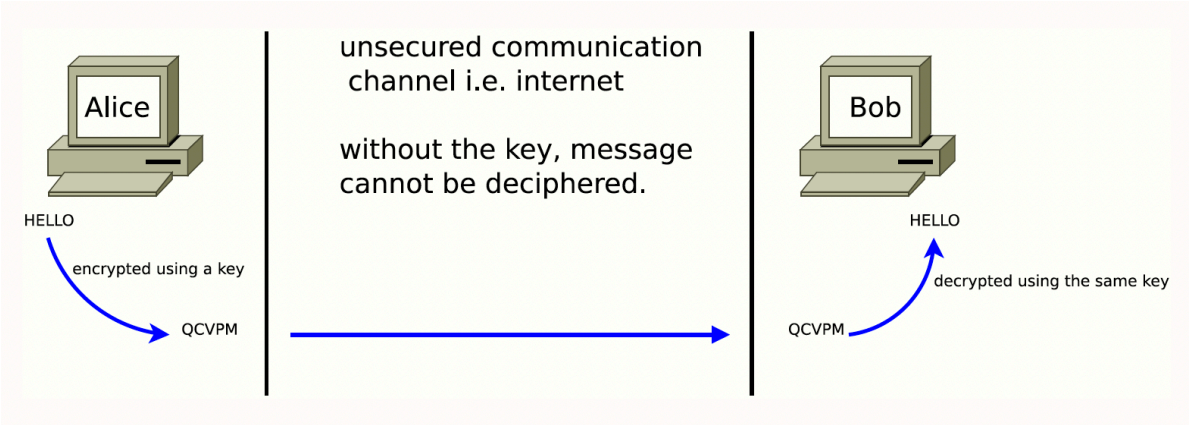

Figure 2: Symmetric-key Cryptography

Common algorithms in this category include **AES** (Advanced Encryption Standard), **3DES**, **Serpent**, **Twofish**, and **Blowfish**.

Symmetric encryption is generally **fast** and computationally inexpensive, making it well-suited for encrypting large amounts of data — video streams, disk volumes, database records, and so on.

> ⚠ The Key Distribution Problem
>
> The fundamental weakness of symmetric encryption is the **key distribution problem**. If Alice and Bob want to communicate securely, they must first agree on a shared key — but how do they share that key securely if they have no secure channel yet? Meeting in person is often impractical, and sending the key over the internet defeats the purpose. This is precisely the problem that asymmetric cryptography was designed to solve.

4.2 Asymmetric Cryptography

Asymmetric cryptography (also called **public-key cryptography**) solves the key distribution problem by using *two* mathematically linked keys: a **public key** and a **private key**. Any message encrypted with one key can only be decrypted with the other — and critically, it is computationally infeasible to derive one key from the other.

The mathematical hardness usually comes from number theory. The most famous example is **factoring large numbers**: it is easy to multiply two large prime numbers together, but extraordinarily difficult to factor the result back into its prime components. This asymmetry — easy one way, hard the other — is the foundation of systems like RSA.

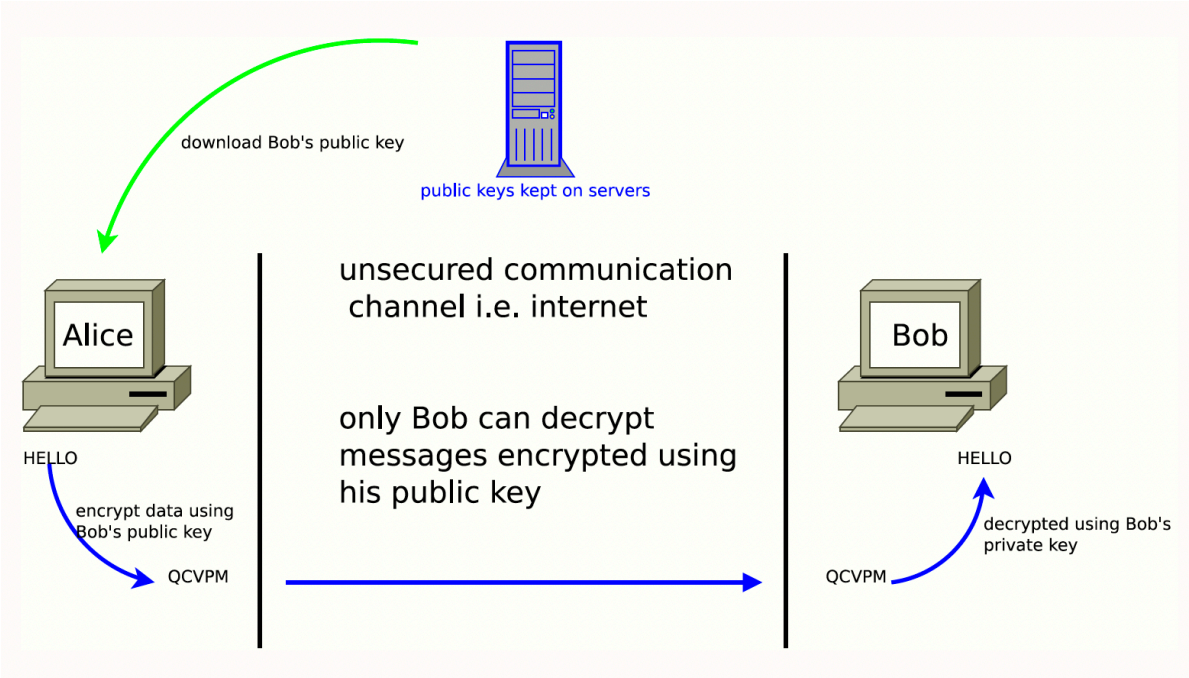

Figure 3: Asymmetric (Public-Key) Cryptography

4.2.1 Confidential Communication

To send Alice a private message, Bob obtains Alice's public key (which she has published openly) and encrypts the message with it. Only Alice's private key can decrypt it — and only Alice has that. Even Bob cannot decrypt his own message after sending it.

This solves key distribution: Alice and Bob never need to have met, and no secret has ever been transmitted over an insecure channel.

4.2.2 Digital Signatures

The public/private key system can also be used in *reverse* to prove identity. If Alice encrypts a message with her *private* key, anyone with her public key can decrypt it. The message is not secret — but everyone who successfully decrypts it knows that only Alice could have created it, since only Alice holds the private key. This is the basis of a **digital signature**.

Digital signatures allow us to answer two critical questions:

- **Is this message from a trusted source?** (authenticity)
- **Has this message been modified in transit?** (integrity)

You interact with this every time you visit an HTTPS website. Your browser verifies the server's digital certificate — signed by a trusted Certificate Authority — to confirm that it is talking to the genuine server and not an imposter.

Common Asymmetric Algorithms

- **RSA** — the classic; security based on integer factorisation
- **Diffie-Hellman** — a key *exchange* protocol (not encryption directly)
- **DSA** (Digital Signature Algorithm) — designed specifically for signatures
- **ECDSA** (Elliptic Curve DSA) — a more efficient variant using elliptic curve mathematics; used in modern TLS certificates and SSH keys

You can see which algorithms are in use by clicking the padlock icon in your browser and inspecting the certificate details.

Symmetric + Asymmetric in Practice

In the real world, these two approaches are almost always used *together*. Asymmetric encryption is powerful but slow — encrypting a large file with RSA directly would be impractical. Instead, modern protocols (like TLS/HTTPS) use asymmetric cryptography to securely exchange a symmetric key at the start of a session, then use that fast symmetric key for the bulk of the data transfer. This hybrid approach gets the best of both worlds.

4.3 Hashing

Hashing is a third application of cryptography that works quite differently from encryption. A hash function takes an input of any size and produces a **fixed-size output** (the *hash* or *digest*) — and crucially, this process is **one-way**: you cannot recover the original input from the hash.

At first this sounds counterproductive. Why scramble a message into gibberish if you can never unscramble it? The answer lies in two properties of a good hash function:

1. **Determinism** — the same input always produces the same hash.
2. **Collision resistance** — it is computationally infeasible to find two different inputs that produce the same hash.

These two properties together make hashes useful for **verification without storage of the original data**.

4.3.1 File Integrity Checking

Imagine downloading a 4 GB Linux ISO on a slow connection. The server publishes the SHA-256 hash of the legitimate file. Once your download completes, you run the same hash function on your local copy — a process that takes only a second or two. If your hash matches the server's published hash, your file is byte-for-byte identical to the original. If they differ, the file was corrupted or tampered with during download.

4.3.2 Password Storage

Storing passwords in plain text is catastrophic when a database is breached — the attacker instantly has every user's password. Instead, systems store only the *hash* of each password. When you log in, the system hashes what you typed and compares it to the stored hash. If they match, you're in — and the original password is never stored anywhere.

When attackers breach a database and obtain password hashes, they cannot directly reverse them. Their best option is a **dictionary attack** or **rainbow table** — pre-computing hashes of millions of common passwords and checking for matches. This is why long, unique, random passwords are important: they are far less likely to appear in an attacker's pre-computed list.

> **Salting**
>
> Modern password storage adds a **salt** — a random value appended to the password before hashing, stored alongside the hash. Even if two users have the same password, their salts differ, so their hashes differ. This defeats pre-computed rainbow tables entirely, since the attacker would have to recompute hashes for every possible salt value.

4.3.3 Common Hash Functions

MD5 converts any input to a 128-bit (32 hex character) hash. Collisions were found as early as 1993, and MD5 is now considered cryptographically broken for security purposes. It is still used for non-security checksums.

```
echo -n "hello world" | md5sum   # note the different hash...
echo "hello world" | md5sum      # ...when a newline is included
```

Try changing even a single character in the input — the resulting hash will look completely different. This is called the **avalanche effect** and is a property of all good hash functions.

SHA (Secure Hash Algorithms) is a family of successively stronger hash functions:

- **SHA-1** — 160-bit output; deprecated since ~2010 due to collision vulnerabilities
- **SHA-256** — 256-bit output; widely used today (TLS certificates, Bitcoin, etc.)
- **SHA-512** — 512-bit output; stronger, used where extra security is needed

```
echo -n "hello world" | sha1sum
echo -n "hello world" | sha256sum
echo -n "hello world" | sha512sum
```

Algorithm	Output Length	Status
MD5	128 bits (32 hex chars)	Broken — use for checksums only
SHA-1	160 bits (40 hex chars)	Deprecated
SHA-256	256 bits (64 hex chars)	Current standard
SHA-512	512 bits (128 hex chars)	Current, higher security

Hashing in Cybersecurity Competitions

Hash cracking and identification are common challenges in CTFs (Capture the Flag competitions) and events like Cyber Storm. You will often be given a hash and asked to find the original value — usually by running a wordlist through the same hash function and looking for a match. Tools like `hashcat` and `john` (John the Ripper) automate this. Recognising which algorithm produced a hash (by its length and character set) is a useful skill.

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5 Summary

This has been a broad tour of cryptography from ancient Sparta to modern HTTPS. The key threads running through all of it are:

- Security is always relative to what an attacker can *compute*. What was secure in 1800 is trivially broken today.
- Modern security comes from **mathematical hardness**, not secrecy of the algorithm. A good system should remain secure even if the attacker knows exactly how it works — the security lives entirely in the key.
- Encryption, signatures, and hashing solve different problems: **confidentiality**, **authenticity**, and **integrity** respectively. Real-world systems like HTTPS use all three together.

Some of these topics will be covered in more detail later in the quarter with hands-on application. Others you will encounter in dedicated courses like CSC 444/544/CYEN 406.[2]

[2]Typically offered every other year in the Spring quarter. Was last offered in Spring 2026.